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The Editors
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Dear Editors,

I am pleased to submit my manuscript, “**Breaking the Observation Ceiling: Conformal Fusion of Trace Signals for Detecting Prompt-Injection Compromise in LLM Agents**,” for consideration as a research article in *Computers & Security*.

Large language model agents that call external tools now read email, browse the web and move money on a user’s behalf. Because they ingest untrusted content and fold it into the context that drives their next action, they are exposed to indirect prompt injection. Runtime monitors that watch only the agent’s black-box execution trace are attractive in practice—they need no model access, no retraining, and can be placed in front of a third-party agent—but the literature has converged almost entirely on a single kind of evidence: deciding whether untrusted text *looks like an instruction*. This paper shows that this choice imposes a hard information limit, and that crossing it requires a different kind of evidence.

The work makes four contributions. First, it introduces **SENTRY-Fuse**, a monitor that converts heterogeneous trace signals—two scoring the observation stream, two scoring the action stream—into conformal tail surprisals against a benign calibration split and fuses them additively. The construction is scale-free, provably order-preserving and free of tuned weights, and the paper proves that a widely-used alternative normalisation is many-to-one and can drive performance to exactly chance, as it did in an earlier version of this work. Second, action-side evidence is shown to be *complementary* rather than redundant: the fused monitor exceeds what payload visibility alone can support, reaching 0.960 ± 0.010 true positives at a 3.66% false-positive rate on a matched corpus, above the strongest published trace-only figure. Third, it characterises the **detectability ceiling** of trace-only monitoring: of 90 genuinely compromised trajectories, 7.8% leave no trace evidence at all, which bounds *every* such monitor and means reported rates above that ceiling are not comparable across corpora unless the corresponding fraction is reported too. Fourth, it documents a label-polarity pitfall in a widely used evaluation harness that inverts the compromised and resisted classes, with the ground-truth procedure that resolves it.

I believe the work suits *Computers & Security* in particular. It is a security paper first: it defines a threat model, evaluates a defence against real attacks on real agent traces, quantifies what that defence can and cannot achieve, and reports its limits rather than only its successes. It also offers two things of direct use to others in the area: a measurement that bounds a whole class of defences, and a methodological warning that affects how results on a standard benchmark should be read. The paper reports its negative results in full, including three signals I implemented and rejected and the finding that a hand-tuned threshold outperformed the anytime-valid machinery from which the project began; these are what allow the positive claims to be interpreted correctly.

All code, the trajectory corpora, the cached model-judge scores and the analysis scripts that regenerate every number, table and figure are publicly released, and the evaluation reproduces offline without access to any model API.

I confirm that this manuscript is original, has not been published previously and is not under consideration elsewhere. There are no competing interests and no external funding was received; as sole author I am responsible for all aspects of the work. Thank you for your consideration, and I look forward to the reviewers’ comments.

Sincerely,

Quang-Vinh Dang